Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

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Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to

their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.
        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider the project below.
        - Create an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the chempy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.

I want my website to function as a way to keep track of plants based on factors such as watering, sunlight, etc. The goal that I am trying to address is that people will be able to use this website to keep track of their plants and keep them alive more effectively.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

I have found a few apps and one free website that works as a journal for keeping track of plants and logging their names, locations, care schedules, etc.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

The main purpose of this site would be for users to input their plants into the site and be able to keep track of their needs. I want to have the user be able to input the needs of the plants such as how much it needs to be watered and how much sun it needs. I also want to

have users be able to check off plants based on if they have been watered yet. Overall, I want this application to be used to be able to keep track of the needs of the user's plants.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The program will take in an input of plants by the user and store them in a list of Plant dataclasses indicating information about them (name, water_frequency, sunlight, and more) and present this to the user so they can keep track of the plants that they input.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The specific information that the user will need to provide to my program would be an input of the name of their plant, how much it needs to be watered, and any other relevant information in an "additional information" section.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The computer will be able to add plants to a list of plants, update plant information, and check whether a plant has been watered.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will see a list of their plants and their care requirements and an indicator of if the plant needs to be watered or not and if it has been yet or not.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

“Plant”, “Water”, “Care Requirements”

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

The most important thing listed above is “Plant”. Details I need: Plant’s name, watering frequency, sunlight needed, if it has been watered Details I will ignore: Height, weight, color

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The part of this project that seems the most difficult or complex might be making sure the plants are updated correctly, especially considering if the user is able to check off if the plant needs water or if the plant has been watered. I also think it might be complex to be able to present all of the data about the plants in a way that makes sense, is clear, and attractive to look at.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will ensure that I understand by looking through each piece of code as if I was debugging

it, making sure that I understand how each part of the code is working. Additionally, if I am having trouble understanding, I can ask the LLM to work through each piece of the code with me and explain more of how exactly it works and what it does.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My ultimate goal for this project is just to make something that me or my friends can use effectively and to make something I am proud about. I am not looking to make anything very impressive or complex, I just want it to actually be functional and be able to help people with their issue (in this case, keeping track of their plants). I do also want to learn more about longer programs through this project and how many functions can work together to make one product.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

“I am creating a website to help people keep track of their plants. What are some common reasons or problems that cause people to struggle with keeping track of their plants and what are some ways to remedy them?”

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

The best ideas the AI suggested were to check off plants that have been watered or display if a plant needs watering to help with those who forget to water their plants, record the different care needs for each plant so that the user does not forget the different needs of the different plants, and include a date for the last time the plant was watered so that the user will not forget how long it has been since the last watering and will know if they need to water their plant yet. It mentioned apps Planta, Greg, and Picture This, as well as just the calendar and notes app for logging dates and information.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

"Sending automatic reminders is too complex for my course because I do not have external services to work with and no way to actually contact the user outside of their general usage of the website. What is a simpler alternative?"

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

Based on this conversation, I am getting more ideas on how I can remind users to take care of their plants. Before, it felt too simple and like a generic list of information about the plants, but now the project feels more focused on reminding the user of the care needs of the plants, which is what I want.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

If I get stuck, I can use an LLM to help me generate some more ideas or help think of other coding aspects to add to my site. Additionally, going to office hours would be a great resource for me to utilize if I get stuck, because I can get help from someone who understands coding but will also be able to get an outside perspective on my site and see how it functions and its quality in the eyes of an observer.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

If my LLM provides code that causes an error or if there is a problem during deployment. I can first look through the code and follow its cycle to see if it is missing anything or if there are any issues with any specific steps. I can then make some unit tests or print statements for different parts of my code to pin-point which area is giving me issues and what is actually working, which will help me go through that section again step-by-step to see what steps are wrong. If I can still not figure out the problem, I can send the error to the LLM that gave me the code and see if they can find the error in their work. If that does not work as well, I can

get help from a professor or TA in office hours to get an extra set of eyes on my program to help debug or find the issue.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I want to make a plank tracking site. The main purpose of this site would be for users to input their plants into the site and be able to keep track of their needs. I want to have the user be able to input the needs of the plants such as how much it needs to be watered and how much sun it needs. I also want to have users be able to check off plants based on if they have been watered yet. Overall, I want this application to be used to be able to keep track of the needs of the user's plants. Can you help me determine the main goal of this site using CS1 concepts. I believe that the input would be the specific information that the user will need to provide to my program would be an input of the name of their plant, how much it needs to be watered, and any other relevant information in an "additional information" section. Can you help me determine the processing and the output?
I think your plant-tracking website is a great idea for a CS1 final project because it solves a real-world problem and can be built using basic programming concepts like variables, lists, conditionals, and functions.

Based on your project description and the IPO Model section of your project worksheet, here's how you can define the main goal, input, processing, and output of your application.
CS1 Final Project Alpha F26.pdf

Part 1: The Core Goal
Main goal: The goal of my plant-tracking website is to help users organize their plants' care requirements and keep track of which plants need to be watered or otherwise cared for.

This describes what the entire program accomplishes for the user in one sentence.

Part 2: Input, Processing, and Output (IPO)
1. Input — What the user provides
The information entered by the user about each plant.

Plant name: e.g., Snake Plant or Aloe Vera.

Watering needs: e.g., every 7 days.

Sunlight requirements: e.g., direct sunlight or indirect sunlight.

Additional information: optional notes about the plant, such as soil preferences or special care instructions.

Watering status: whether the plant has been watered or still needs watering.

2. Processing — What the program does
The program organizes and updates the plant information entered by the user.

Store each plant's information in a list or another suitable data structure.

Add new plants to the user's plant collection.

Check whether a plant has been marked as watered.

Update a plant's watering status when the user checks a box.

Organize plants so users can identify which ones still need attention.

Save plant information in browser storage so it can remain available when the user returns.

3. Output — What the user sees
The program displays an organized plant-care dashboard based on the information entered and processed.

A list of the user's plants and their care requirements.

Each plant's watering frequency and sunlight needs.

A checkbox or button showing whether the plant has been watered.

A visual indicator of which plants still need watering.

The updated plant list whenever the user adds a plant or changes its status.

Part 3: How the IPO model works together
Here's a simple example of how a user would interact with your site.

INPUT

Plant: Aloe Vera

Watering: Every 7 days

Sunlight: Bright, indirect light

The program stores the information and checks the watering status.

OUTPUT — PLANT DASHBOARD

Shop Pennington Aloe Vera with 4-in W x 4-in H Terracotta Clay Planter and Saucer at Lowes.com

Aloe Vera

Water every 7 days

Bright, indirect light

Needs watering

After the user checks the watered box, the program updates the plant's status and displays it as watered.

Part 4: Connecting your project to CS1 concepts
Your idea naturally uses several introductory programming concepts.

CS1 concept

How you would use it

Variables

Store a plant's name, watering frequency, and sunlight needs.

Lists

Keep track of multiple plants in one collection.

Dictionaries or data structures

Group each plant's information together, if supported by your course.

Conditionals (if / else)

Determine whether a plant is marked as watered.

Functions

Create reusable actions to add a plant, update its status, or display the plant list.

Loops

Go through the plant list to display every plant or find plants needing attention.

Boolean values

Represent watering status as True or False.

Local storage

Preserve plant information between visits using the browser's storage features supported by Drafter.

Part 5: Keep the project manageable
For a first programming project, I recommend starting with three essential features:

Add a plant with a name, watering frequency, sunlight requirements, and optional notes.

View all plants in an organized dashboard.

Check off watering and update each plant's status.

You could add reminders based on the last watering date later, but that feature requires storing dates and calculating when a plant is due. It is not necessary for the first version.

One important distinction: a checkbox can record whether a plant has been watered, but automatically determining whether it is due for watering requires additional information, such as the last watering date. You can make that a later improvement if you have time.

Since Drafter is a frontend Python library, browser local storage is a realistic way to keep the plant list between visits without needing to build a server or database.

Your next step: Use these four IPO sections to fill in Part 2 of your worksheet, then identify your main data object for Part 3: a Plant with properties such as name, watering frequency, sunlight needs, and watering status.

What parts of this project seem the most difficult or technically complex?
For your plant-tracking website, the most difficult parts will likely be organizing each plant's information, updating watering statuses, and saving the information so it is still available when the user returns. These challenges are manageable using CS1 concepts.

Here is a response you can use for Part 4, Question 1 of your project worksheet.

The most difficult parts of this project will likely be storing and organizing information for

multiple plants, updating each plant's watering status when the user checks it off, and saving the information so that it remains available when the user returns to the website. I will need to make sure that each plant's information, such as its name, watering needs, and sunlight requirements, stays organized and is updated correctly. Another challenge may be designing the website so that users can easily add plants and view their care information. I plan to focus on the basic features first and use CS1 concepts such as lists, variables, conditionals, and functions to make the program manageable.

Why these parts may be challenging
Organizing plant data: Keeping the correct information connected to each plant when users add multiple plants.

Updating checkboxes: Making sure that checking off one plant as watered updates that plant without accidentally changing another plant's status.

Saving information: Using browser local storage so plants do not disappear when the page is refreshed or reopened.

Displaying the information: Designing a clear dashboard that makes it easy to see each plant's care requirements.

My recommendation: Focus on adding plants, displaying their information, and checking off watering first. Saving the data between visits can be your next priority. Automatic watering reminders based on dates can be an optional extension if you have time.

This might also interest you.

I am creating a website to help people keep track of their plants. What are some common reasons or problems that cause people to struggle with keeping track of their plants and what are some ways to remedy them? Do not write any code.
For your plant-tracking website, a good place to start is understanding that people often struggle with plant care because each plant has different needs, it can be difficult to remember care schedules, and people may not know when they last watered a plant. Your website can help solve these problems by keeping plant information organized in one place.

Common plant-care problems and possible solutions
25 Best Indoor Plants to Brighten Any Room

1. Forgetting to water plants

People get busy with school, work, or everyday responsibilities and forget which plants need watering.

Possible solution

Let users check off plants after watering them and display which plants still need attention.

Relax & Refresh | Plant Bundle
| 3 X PLANTS | 3 X POTS

2. Different plants have different needs

Some plants need frequent watering, while others prefer their soil to dry out between waterings. Sunlight requirements also vary.

Possible solution

Allow users to record each plant's watering frequency, sunlight requirements, and additional care instructions.

The Power of Plants- How Indoor Gardening Can Improve Your Mental Health

3. Forgetting when a plant was last watered

A person might remember watering a plant recently but not know whether it was yesterday or several days ago.

Possible solution

Include a last-watered date and a simple indicator of whether a plant may need attention soon.

Plant Care Life Tips: Discreet Pot Rim Labeling

4. Losing track of plant information

Care instructions may be scattered across websites, notes, labels, or memory.

Possible solution

Provide one organized dashboard where users can view each plant's name, care needs, and personal notes.

Easy Harvesting — Advanced Plant Care, Made Simple

5. Overwatering or underwatering

People may water plants too often or too infrequently because they follow the same schedule for every plant.

Possible solution

Display each plant's preferred watering frequency and care notes. Explain that the schedule is a guideline, since soil moisture, temperature, and season can affect watering needs.

Which problems should your website focus on first?
Since this is a CS1 project, I recommend choosing features that solve the biggest problems without making the program too complicated.

Priority

Feature

CS1 concept involved

Essential

Add and view plants

Variables, lists, functions

Essential

Check off plants after watering

Booleans, conditionals

Essential

Store watering and sunlight needs

Data organization

Useful

Record the last-watered date

Dates and data updates

Optional

Show plants that are due for watering

Date comparisons and conditionals

Optional

Send automatic reminders

More complex browser features or external services

The biggest opportunity for your website is to combine a plant's care instructions with a simple tracking system. Rather than just telling users how to care for plants, the website would help them remember what they have already done.

For your project worksheet, you could summarize your findings like this:

Some common problems people experience when caring for plants are forgetting to water them, not knowing how much sunlight or water each plant needs, forgetting when they last watered a plant, and having trouble keeping track of multiple plants. My website could help solve these problems by allowing users to store each plant's care requirements, view all their plants in one place, and check off plants after watering them. In the future, I could add a last-watered date and indicators to help users identify which plants may need attention. These features would make plant care more organized and easier to manage.

For building a plant-tracking app, software prototyping tools can help.

What are some existing solutions or prior apps that help with these problems?
Several existing apps already help people care for their plants. For your project worksheet, the most relevant examples are Planta, Greg, and simple calendar or notes apps because they address watering schedules, plant-care information, and keeping track of completed tasks.

Existing solutions and prior apps
Apple välkomnar sin 59e svenska utvecklare till App Store Foundations Program - Apple (SE)

1. Planta

Best for plant-care schedules

Planta lets users organize their plants, receive watering and other care reminders, and access information about plant needs. It also offers plant identification and light-measuring features.

App Store
+1

Connection to your idea: Your site could offer a simpler plant list, care requirements, and a checkbox to mark watering as complete.

Visit Planta's website

Greg App Page

– Costa Farms

2. Greg

Best for watering guidance and community

Greg helps users track plant care and watering, with plant identification and features for getting advice from other plant owners.

Growli
+1

Connection to your idea: You could focus on a straightforward tracking experience without needing identification technology or social features.

Visit Greg's website

I’m no longer a plant killer after I started using the ‘PictureThis’ app

3. PictureThis

Best for identifying plants

PictureThis identifies plants from photos and provides plant-care information and tools for assessing plant health.

Planterra Journal
+1

Connection to your idea: Your site could let users manually enter plant names and their care needs instead of building photo-recognition features, which would be much more complicated for a CS1 project.

Visit PictureThis's website

Terragrix

4. Calendar apps and notes apps

Best for simple, general-purpose tracking

Users can create recurring calendar reminders for watering and keep plant names or care instructions in notes. These tools aren't specifically designed for plant care, but they can help people remember tasks.

Sow
+1

Connection to your idea: Your website could combine reminders or care schedules with a dedicated list of plants and their individual needs.

What could make your website different?
You don't need to invent an entirely new type of application. Your goal can be to create a simpler, beginner-friendly alternative that focuses on the core problem.

Keep it simple: Avoid overwhelming users with advanced features.

Make watering status easy to update: Use a checkbox for each plant.

Keep care information together: Display watering frequency, sunlight needs, and notes on each plant.

Focus on tracking rather than identification: Let users enter their plant names themselves instead of implementing photo recognition.

Worksheet-ready response
Some existing solutions for plant care include Planta, Greg, PictureThis, and general-purpose calendar or notes apps. Planta helps users organize plants and receive watering reminders, while Greg provides plant-care tracking and watering guidance. PictureThis focuses on identifying plants and providing care information. People can also use calendar apps to set reminders or notes apps to store plant information. My website would provide a simpler way to manage plant care by allowing users to enter their plants' names, record their watering and sunlight requirements, view all their plants in one place, and check off plants after watering them. This would help users stay organized without requiring complicated features such as photo identification or automated plant diagnosis.

For building a plant-tracking app, here’s a no-code AI builder option.

Sending automatic reminders is too complex for my course because I do not have external services to work with and no way to actually contact the user outside of their general usage of the website. What is a simpler alternative?
A simpler alternative would be to create an on-screen watering status dashboard instead of sending automatic reminders. This keeps the project within the scope of a CS1 course because everything happens while the user is using your website.

Simpler alternatives to automatic reminders
1. On-screen care dashboard (recommended)

Display all the user's plants with a status such as "Needs watering" or "Watered today." Users can see what needs attention as soon as they open the website.

Easy to implement
2. Last-watered date

Let users record the date they last watered each plant. The website can display that date so users can decide whether it needs watering again.

Moderate difficulty
3. Daily care checklist

Show a checklist of plants the user wants to check on. Users can mark each plant as watered and reset the checklist when they begin a new day or care session.

Easy to implement
My recommendation for your project
Combine options 1 and 3. Your homepage could show a plant list like this:

My Plant Care Dashboard
Example of how your website might look

Snake Plant

Water every 2–3 weeks

Needs attention

Watered
Peace Lily

Check soil regularly

Marked as watered

Watered
Aloe Vera

Water when soil is dry

Needs attention

Watered
Interactive mockup: try checking a box to see how the status changes.

This approach lets you practice important CS1 concepts:

Lists: Store multiple plants.

Conditionals: Display different messages depending on watering status.

Functions: Update a plant's status when a checkbox is selected.

Local storage: Preserve the plant list and status between visits.

One important detail: the dashboard should help users track care rather than claim a plant definitely needs water based only on its schedule. Actual watering needs can change with soil moisture and growing conditions.

Worksheet-ready response:

Automatic reminders would be too complex for this project because they require external services or ways to contact users outside the website. A simpler alternative is to create an on-screen plant-care dashboard that displays each plant's watering requirements and current watering status. Users can check off plants after watering them and record when they last watered each plant. This approach still helps users stay organized while using basic CS1 concepts such as lists, conditionals, functions, and variables.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.